

Module :9 Infrastructure services –

8- Host A and Host B sit in two different subnets. The path between the subnets of these two hosts runs through three different Layer 3 forwarding devices (routers and Layer 3 switches). A network engineer uses the APIC-EM Path Trace ACL Analysis tool to analyze the path used for Host A to send packets to Host B.

Which part of the function is done specifically by the ACL Analysis or ACL Trace part of the tool?

- A. Discovery of the topology that exists between the two hosts
- B. Analysis of the Layer 3 forwarding decisions in the path from Host A to B
- C. Analysis of the Layer 2 forwarding decisions in the path from Host A to B
- D. Analysis of the impact of ACLs on the packets that would flow from Host A to B**

ANS- The ACL Analysis (ACL Trace) part of the APIC-EM Path Trace tool is used to check how Access Control Lists (ACLs) affect packets traveling from Host A to Host B. It analyses which ACL rule matches the packet and whether the packet is permitted or denied on each Layer 3 device. It does not focus on Layer 2 forwarding decisions or only discovering the topology. Its main purpose is to examine the effect of ACL rules on the packet flow

HOST A → 10.0.0.1

Host B → 10.0.0.2

1.0.0.0.1 → router1 → router 2 → 20.0.0.1

On Router 1 ACL

- Access-list 101 deny Tcp host 10.0.0.1 host 20.0.0.2 eq 80
- Do WR
- Access-list permit 101 ip any any
- Do wr

Deny HTTP

9- Which IPv6 address is the equivalent of the IPv4 interface loopback address 127.0.0.1?

A. ::1

B.:

C. 2000: :/3

D. 0: :/10

ANS- In IPv4, 10.0.0.1 is the loopback address. It is used to test the local device. In IPv6, the equivalent loopback address is ::1. The address ::1 always refers to the local device itself and is mainly used for testing and troubleshooting purposes.

- IPv4 10.0.0.1 (IPv4 loopback)
- Ipv6::1 (IPv6 loopback)

If you type ping :: 1 on a router and get reply, it means ipv6 is working correctly.

10- Which command is used to apply an ACL to an interface?

A. access-group

B. ip access-group

C. ip access-list

D. ip access-class E. access-class F. access-list

ANS- To apply an ACL to an interface in Cisco IOS, the command ip access-group is used in interface configuration mode. First, the ACL is created using access-list or ip access-list. After creating it, the ACL must be applied to the interface using: ip access-group ACL_NAME in

This command activates the ACL on the interface to filter incoming or outgoing traffic.

Applying ACL to interface (10.0.0.0 series example)

STEP 1 CREATE ACL

Block 10.0.0.0 network:

- Access-list 105 deny Tcp host 10.0.0.0
- Do WR
- Access-list permit 105 ip any any
- Do WR

Block Ips from 10.0.0.1 to 10.0.0.254

Allow others

- Int se0/0
 - Ip access-group 105 in
 - No shut
 - Do wr
 - Int fa0/0
 - Ip access- group 105 out
 - No shut
 - Exit
 - Do wr
- 10.0.0.1 blocked
10.0.0.2 blocked
30.0.0.1 allowed

11- Which command and mode will successfully configure a hostname of R1 on a `Cisco IOS router?

A. Router(config)#name R1

B. Router# hostname R1

C. Router(config)#hostname R1

D. Router#name R1 E. Router>hostname R1 F. Router>name R1

ANS- To change the hostname of a Cisco router, the command hostname R1 must be entered in global configuration mode.

After entering configure terminal, the router enters global configuration mode, and then the hostname can be changed successfully.

Before:

enable

Router# configure terminal

Router(config)# hostname R1

After:

R1(config)#

Hostname successfully changed.

12- Which of the following reserved IPv4 addresses has binary 0s in all of the host bit positions?

A. Local broadcast address

B. Loopback address

C. Directed broadcast address

D. Network address

E. All zeros address

ANS- In IPv4 addressing, when all the host bits are 0, the address is called the network address. The network address identifies the entire network and cannot be assigned to any device. For example, in the network 10.0.0.0 , the address 10.0.0.0 is the network address.

Network:

10.0.0.0/24

Address range:

- 10.0.0.0 → Network address (cannot assign)
- 10.0.0.1 → First usable host ✗
- 10.0.0.2 → Host ✓
- 10.0.0.254 → Last usable host ✓
- 10.0.0.255 → Broadcast address ✗

Host bits are all 0 in 10.0.0.0, so it is the network address.